



CBS
ENGINEERING

Company Profile

2025

Table of Contents

About Us	03
Our Services	04
Where We Work	05
Why Choose Us?	06
Electrical Projects	07
Solar and Power Projects	13
Interior Flnishing Projects	19

About Us

Cangasir Building Services Engineering Limited (CBS) is a Zambian construction company established in 2024. We provide specialized services in electrical installations, plumbing systems, HVAC solutions, and building finishing for residential, commercial, and industrial projects. Our objective is to deliver reliable, cost-effective, and compliant solutions that align with industry standards and client requirements.



At CBS, we don't just build systems. We build peace of mind.



Our team of certified professionals brings together years of technical expertise, precision workmanship, and a commitment to safety and sustainability. From rough-in to final finish, we handle every phase of your building systems with care and coordination, ensuring smooth project execution and lasting performance.

At CBS, we believe that strong client relationships are built on transparency, responsiveness, and consistent delivery. Whether it's new construction or an upgrade to an existing facility, CBS offers dependable services designed to meet your project's technical and operational needs.



Our Mission

To provide dependable, high-quality building services that enhance and empower living and working spaces. .



Our Vision

To be the preferred choice in the construction services industry by consistently delivering innovative, reliable, and high-quality building solutions.

Our Services



Electrical

Electrical Services

Electrical system installation and maintenance

Wiring and rewiring

Lighting solutions (interior, exterior, emergency)

Energy-efficient electrical upgrades

Panel upgrades and circuit breaker installations



Plumbing

Plumbing Services

Residential and commercial plumbing installations

Leak detection and pipe repairs

Water heating systems installation and servicing

Drainage and sewage system maintenance

Gas piping and appliance installation



HVAC

HVAC Services

Heating, ventilation, and air conditioning system installation

HVAC system repairs and maintenance

Ductwork design and installation

Indoor air quality solutions

Smart HVAC system integration



Finishing

Building Completion & Finishing

Interior and exterior painting

Flooring installation (tiles, hardwood, laminate, etc)

Drywall installation and finishing

Ceiling works (suspended ceilings, plastering, etc.)

Carpentry and joinery work

Where We Work

CBS provides services across Zambia, and supports various projects in both urban and rural areas.



Why Choose Us?

Selecting a reliable partner for your building services can be challenging—especially when quality, timelines, and safety are on the line. We understand the pressure that comes with managing construction projects and the importance of working with a team that delivers on its promises.

Here's why CBS Engineering is a trusted choice for building services in Zambia



Experienced Professionals

Our team includes certified electricians, plumbers, HVAC technicians, and finishing specialists.



Innovation

We embrace modern technologies and methods to enhance efficiency, sustainability, and project value.



Customer-Centric Approach

Your satisfaction is our priority, and we provide personalized services to meet your unique needs.



Safety & Compliance

All our work complies with building codes and safety regulations, providing peace of mind for clients.



On-Time Delivery

We deliver projects on time, ensuring efficiency while upholding quality.



Quality Workmanship

We maintain top industry standards and use premium materials for durability and performance.

Electrical Projects

Our electrical projects demonstrate our capability in delivering safe, efficient, and compliant systems. From power distribution to lighting and backup systems, we ensure every installation meets the highest technical standards.

PROJECT

Provided design, consultation, and quality control services for the wiring and installation works.

Client

[Client Name]

Site

Chelston, Lusaka

References

Contact Person

[Role]

Contact Person

[Role]

Scope of Work

- ✓ Interior and exterior lighting system
- ✓ Ventilation, air conditioning and air quality system
- ✓ 16 Channel PoE NVR CCTV system.
- ✓ Sound and Public Address (PA) system.
- ✓ LT switchgear, power points and power distribution system.
- ✓ Generator and UPS power backup system.
- ✓ TV and projector display systems.
- ✓ Fire detection system.
- ✓ Technical drawings, HVAC PID and electrical single line diagrams.

PROJECT

Consultation and installation of five (5) 37kW pump and motor assembly systems across two pump stations.

Client

[Client Name]

References

Contact Person

[Role]

Site

Mufulira (multiple locations)

Contact Person

[Role]

Scope of Work

- ✓ Installation of five (5) 37kW pump and motor assembly systems for two pump stations.
- ✓ Installed the interior lighting, security lighting and emergency lighting.
- ✓ Installed single and three phase power points in the pump houses.
- ✓ Installed the DC clips, 3x35mm GI strip and 2 metre earth rods for the lightning protection system for the pump houses.
- ✓ Installed and calibrated electronic ultrasonic level sensors in surge tanks, hydrostatic level sensors in reservoir tanks and overhead tanks, pressure sensors in the pipes for the SCADA system.
- ✓ Ran 4core armoured cables from the distribution substation to the pump houses as the power incomer via trenches.
- ✓ Designed the distribution feeder pillar for the two pump stations.
- ✓ Installed two (2) 3 phase 10kVA Schneider Uninterrupted power supplies. (UPS).
- ✓ Installed five (5) data logger for the electromagnetic flow for the SCADA and online monitoring system.
- ✓ Installed a prewired freestanding Schneider electric panel consisting of scada system, electromechanical controls and soft starters for the water pumps.
- ✓ Installed a conventional fire alarm panel system including smoke detectors, fire detectors, LPG detectors, fire alarms and manual call points.

- ✓ Liaised with ZESCO in the application for and connection of permanent power to the pump stations.
- ✓ Wired a 3 tonne and a 2 tonne electric overhead crane.
- ✓ Installed five (5) electromagnetic flow meters in the and two (2) electronic flow meters in the water network.
- ✓ Installed power and control cables for 2 pump stations.
- ✓ Successfully tested, commissioned, created as built drawings and trained Mulonga water pump attendants, electricians and staff.
- ✓ Installed and calibrated seven (7) electric butterfly valves.
- ✓ Installed two (2) 16kVar 3 phase power correction units (PCU)/ capacitor banks for the motors.

PROJECT

Designed the Heating, Ventilation, and Air Conditioning (HVAC) systems for a proposed church building

Client

[Client Name]

References

Contact Person

[Role]

Site

Lusaka

Scope of Work

- ✓ 204,000 btu/h cooling capacity, 3-pipe Multi-split variable refrigerant flow (VRF) system for the church offices.
- ✓ Direct expansion (DX), 926,000 btu/h cooling capacity air conditioning system for the church hall.
- ✓ Fresh air ventilation system with heat recovery for the church hall.
- ✓ Air duct system with filters, dampers and fire dampers for quality air distribution.
- ✓ Technical drawings, HVAC PID and electrical single line diagrams.

PROJECT

Designed electrical and mechanical systems for the proposed classrooms, offices, and multipurpose hall in Lusaka

Client

[Client Name]

References

Contact Person

[Role]

Site

Long Acres, Lusaka

Scope of Work

- ✓ Interior and exterior lighting system
- ✓ Ventilation, air conditioning and air quality system.
- ✓ LT switchgear, power points and power distribution system.
- ✓ Generator and UPS power backup system.
- ✓ Fire detection system.
- ✓ Technical drawings, HVAC PID and electrical single line diagrams.

PROJECT

Design and engineering services for the proposed eight (8) flat apartments in Roma, Lusaka

Site

Roma, Lusaka

References

Contact Person (also client)

[Role]

Scope of Work

- ✓ Interior, exterior, and security lighting systems
- ✓ Air conditioning and air quality systems
- ✓ LT switchgear, power points, and power distribution systems
- ✓ Fire detection system
- ✓ Technical drawings, including electrical single-line diagrams

PROJECT

Design and installation of a complete CCTV surveillance system

Client

[Client name]

References

Contact Person

Site

Meanwood Ibex, Lusaka

Scope of Work

- ✓ Successfully designed and installed a 12-channel analog CCTV system.
- ✓ Installed 3tb HDD for the DVR
- ✓ Installed 4x 2.0MP bullet cameras
- ✓ Installed 6x 4.0MP IR bullet cameras.
- ✓ Run the Siamese cables for the cameras to the DVR and the power supply.
- ✓ Created Autocad technical drawings to minimize blind spots for the CCTV (field of view) FoV
- ✓ Successfully tested, commissioned the system and trained occupants on the operation and maintenance of the system

PROJECT

Completed the electrical design and wiring for three residential flats in Meanwood Vorna Valley

Site

Meanwood Vorna Valley, Lusaka

References

Contact Person (also client)

Scope of Work

- ✓ Designed and installed the interior and exterior lighting system.
- ✓ Designed and installed the power points, cooker control unit and geyser switches for the flats
- ✓ Designed and installed the distribution board for each flat
- ✓ Liaised with ZESCO in the application and connection of permanent power
- ✓ Development of technical drawings, bills of quantities (BoQs), and electrical single-line diagrams

PROJECT

Delivered electrical design and wiring services for three residential flats in Lusaka West

Site

Lusaka West, Lusaka

References

Contact Person (also client)

Scope of Work

- ✓ Designed and installed the interior and exterior lighting system.
- ✓ Designed and installed the power points, cooker control unit and geyser switches for the flats.
- ✓ Designed and installed the distribution board for each flat
- ✓ Liaised with zesco in the application and connection of permanent power.
- ✓ Development of technical drawings, bills of quantities (BoQs), and electrical single-line diagrams

Solar and Power Projects

We deliver reliable solar and backup power systems ensuring energy independence and uninterrupted supply. Our work combines technical expertise with clean energy innovation, built to suit both homes and businesses.

PROJECT

Design, consultation and quality control of the wiring, installation and maintenance of a 471kWp grid-tied solar system

Client

[Client Name]

Site

Long Acres, Lusaka

References

Contact Person

[Role]

Contact Person

[Role]

Scope of Work

- ✓ Design and installation of 1248 450W roof mounted monocrystalline solar panels.
- ✓ Installation of 13 3-phase paralleled 27.6kW AC SolarEdge inverters.
- ✓ Installation of 624 Solar optimizers.
- ✓ Running 76 750Vdc strings from the solar panels to the solar system substation.
- ✓ Installation of online monitoring, PLC and SCADA system for the solar power plant.
- ✓ Installation of the solar panel lightning protection system
- ✓ Installation of 1000Vdc rated DC combiner panell
- ✓ Installation 500Vac combiner panel
- ✓ Created technical drawings and electrical single line diagrams

PROJECT

Design, consultation and quality control of the wiring, installation and maintenance of a standalone ground mounted 12kWp solar system

Client

[Client Name]

Site

Chifunabuli, Luapula

References

Contact Person

[Role]

Contact Person

[Role]

Scope of Work

- ✓ Design, assembly and installation of ground mounting structure and 12 monocrystalline solar panels.
- ✓ Installation of 1x12kWac 3-phase Deye inverters.
- ✓ Installation of 15kWH 96V FreedomWon LiFeO4 Battery.
- ✓ Running 3 200Vdc strings from the solar panels to the solar system DC switchboard.
- ✓ Installation and setup of the online monitoring, dongle and the battery-inverter communication system.
- ✓ Installation of 200Vdc rated DC switchboard and surge protection.
- ✓ Installation 400Vac distribution board.
- ✓ Installation of the solar panel lightning protection system.
- ✓ Created technical drawings and electrical single line diagrams.

PROJECT

Designed the technical and financial model of a turnkey 1.5MWp standalone solar system with a battery energy storage system for a shopping mall.

Client

[Client Name]

References

Contact Person

Site

Lusaka West, Lusaka

Scope of Work

- ✓ Designed and simulated PV system to meet the energy requirements for the mall using Helioscope.
 - ✓ Designed an 8MWH Battery Energy Storage System (BESS).
 - ✓ Designed the 6 bay distribution substation with electric meters for each shop.
 - ✓ Created an economised schedule and scope of works.
 - ✓ Designed the ground mounting structure, DC combiners, AC switchboard, lightning protection system and cable trenches.
 - ✓ Created technical drawings and electrical single line diagrams.
-

PROJECT

Installed a residential power backup system in Chipata, ensuring uninterrupted electricity for essential household loads.

Site

Mutenguleni, Chipata

References

Contact Person (also client)

Scope of Work

- ✓ Installed a 1.6kVa power backup system.
- ✓ Installed a 1.6kVA transformer based inverter.
- ✓ Installed 2x 100AH gel 12V batteries.
- ✓ Integrated the power backup system and the grid.
- ✓ Isolated the essential loads in the house from the non-essential loads to accommodate the inverter rating.
- ✓ Trained the occupants of the house on the power backup system.

PROJECT

Designed and installed a complete hybrid solar power system for a residential property

Site

Lusaka West, Lusaka

References

Contact Person (also client)

Scope of Work

- ✓ Designed a 6.6kWp hybrid solar system complete with battery backup system
- ✓ Installation of single phase 10 kW Deye inverter.
- ✓ Installation of 100Ah x2 (10kWh) Dyness lithium batteries.
- ✓ Installation of 550W x 12 Jinko solar panels.
- ✓ Trained the occupants of the house on the solar system.

PROJECT

Designed and installed a hybrid solar power system with integrated battery backup for a residential property

Site

Avondale, Lusaka

References

Contact Person (also client)

Scope of Work

- ✓ Designed a 4.4kWp hybrid solar system complete with battery backup system.
- ✓ Installation of 10 kW Sun Energy inverter.
- ✓ Installation of 200Ah x1 (10kWh) Dyness Lithium battery
- ✓ Installation of 550W x8 Jinko solar panels.
- ✓ Trained the occupants of the house on the solar system.

PROJECT

Designed and installed a hybrid solar power system for a residential property

Site

Chalala, Lusaka

References

Contact Person (also client)

Scope of Work

- ✓ Designed a 5kWh battery power backup system.
- ✓ Installation of 8kW Luxpower hybrid inverter.
- ✓ Installation of 100Ah x1 (5kWh) Dyness Lithium battery
- ✓ Trained the occupants of the house on the power backup system

PROJECT

Designed and installed a hybrid solar power system with integrated battery backup for a residential property

Site

Meanwood Vorna Valley, Lusaka

References

Contact Person (also client)

Scope of Work

- ✓ Designed a 2.2kWp hybrid solar system complete with battery backup system
- ✓ Installation of 5kVA Must hybrid inverter
- ✓ Installation of 100Ah x1 (5kWh) Infinisolar Lithium battery
- ✓ Installation of 550W x4 Jinko solar panels
- ✓ Trained the occupants of the house on the solar system

PROJECT

Designed and installed a hybrid solar power system for a residential property

Site

6 miles, Lusaka

References

Contact Person (also client)

Scope of Work

- ✓ Designed a 2.2kWp standalone solar system complete with battery backup system
- ✓ Installation of 5.5kVA hybrid inverter
- ✓ Installation of 100Ah x1 (5kWh) Lithium battery
- ✓ Installation of 550W x4 Trina solar panels

Interior Finishing Projects

Our interior finishing projects combine thoughtful design with skilled execution to deliver refined, high-quality spaces. From custom ceiling structures to final paint, we ensure that every detail contributes to a cohesive and functional interior environment.

PROJECT

Design, consultation and quality control of the wiring, installation and maintenance of a 471kWp grid-tied solar system

Client

[Client Name]

Site

Chelston, Lusaka

References

Contact Person

[Role]

Contact Person

[Role]

Scope of Work

- ✓ Design of the ceiling board
- ✓ Installation of ceiling structural grid
- ✓ Installation of gypsum board
- ✓ Create and align cut-outs for the light fittings, sockets, speakers and ceiling fans
- ✓ Skimming of the ceiling board
- ✓ Painting of the ceiling board

Building. Excellence.





FACEBOOK
CBS Engineering Limited